Assignment: Query Processing
CSE 511: Data Processing at Scale

Introduction

In this assignment, you will implement a simplified **query processor** that accesses data from **partitioned Reddit rating tables** (created in Assignment: Data Fragmentation). You will extend your prior work by implementing two query types: **Range Query** and **Point Query**.

Step 0: Environment Setup (0 pts)

Option A: Local Environment of PostgreSQL 14 (**Assignment: Data Fragmentation**). All related files can be downloaded here.

Option B: Docker image here. The image already contains the fragmentation database and populated subreddits, range_part, and rrobin_part tables. You do not need to run your Assignment 2 fragmentation.py or tester.py. PostgreSQL starts automatically when the container starts.

```
docker pull change1472/cse511-query-processing:latest
docker run --name cse511-query-processing -it change1472/cse511-query-processing:latest
python3 tester.py
```

- **query_processing.py** — your implementation.
- **tester.py** — the grading/test scripts.

Step 1: Range Query (50 pts)

Function to Implement:

`range_query(parent_partition_table_name, utc_min_val, utc_max_val, save_table_name, connection)`

Note:

1. Use the **parent partition table** (range_part or rrobin_part; both are preloaded in the Docker image).
2. The saved table must contain all tuples for which the `created_utc` value is larger than `utc_min_val` and less than or equal to `utc_max_val`. i.e. (utc_min_val, utc_max_val].
3. The Python function does not need to return rows. Store all matching rows and columns in a new table named `save_table_name`.
4. PostgreSQL does not guarantee row order when reading a table without ORDER BY. To display results in ascending created_utc order, use ORDER BY created_utc ASC when retrieving the saved table. Grading compares saved rows independent of physical order.

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Step 2: Point Query (50 pts)

Function to Implement:
`point_query(parent_partition_table_name, utc_val, save_table_name, connection)`
Note:

1. Use the **parent partition table**
2. Retrieve all tuples where: `created_utc=utc_val`
3. The Python function does not need to return rows. Store all matching rows and columns in a new table named `save_table_name`.

Submission Requirements & Guidelines

- **Individual work:** This is an individual assignment; all work must be your own.
- **What to submit on Canvas:** Submit **exactly one** Python file named `query_processing.py`.
- **Filename/extra files penalty:** Using a different filename or submitting additional files will incur a **30-point penalty**.

Submission Policies

- Late submissions will *absolutely not* be graded (unless you have verifiable proof of emergency). It is much better to submit partial work on time and get partial credit than to submit late for no credit.
- Every student needs to *work independently* on this exercise. We encourage high-level discussions among students to help each other understand the concepts and principles. However, a code-level discussion is prohibited, and plagiarism will directly lead to the failure of this course. We will use anti-plagiarism tools to detect violations of this policy.

Grading Rubric (100 pts)

Part 1: Range Query (50 pts)

- Ten tests, 5 pts each: five use range_part and five use rrobin_part.

Part 2: Point Query (50 pts)

- Ten tests, 5 pts each: five use range_part and five use rrobin_part.

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```
================================================================================
chang@chang-pc:~/docker-file/query_processing$ python3 tester.py
================================================================================
ASSIGNMENT: QUERY PROCESSING - GRADING RUBRIC
================================================================================
Total Points: 100
Step 1: Point Query (50 points)
  - Range Partitioned Table Tests (25 points)
  - Round-Robin Partitioned Table Tests (25 points)
Step 2: Range Query (50 points)
  - Range Partitioned Table Tests (25 points)
  - Round-Robin Partitioned Table Tests (25 points)
================================================================================

-----------------------------Starting Tests---------------------------------
+ Using existing fragmentation database with partitioned tables

+ Cleaning up previous test results...
+ Cleanup complete

================================================================================
STEP 1: POINT QUERY TESTS (50 points)
================================================================================

--- Point Query on Range Partitioned Table (25 points) ---

[Test 1] Basic point query with existing UTC value
Query results saved to table: Test1
        ✓ Test 1 - Basic point query: PASSED (5 points)

[Test 2] Point query with different UTC value
Query results saved to table: Test2
        ✓ Test 2 - Different UTC value: PASSED (5 points)

[Test 3] Verify point query creates table successfully
Query results saved to table: Test3
        ✓ Test 3: PASSED (5 points) - Table created successfully

[Test 4] Verify results are stored in correct table format
Query results saved to table: Test4
        ✓ Test 4: PASSED (5 points) - Table created with 1 rows and 15 columns

[Test 5] Verify parent partition table is used correctly
Query results saved to table: Test5
        ✓ Test 5: PASSED (5 points) - Parent partition table used correctly

--- Point Query on Round-Robin Partitioned Table (25 points) ---

[Test 6] Basic point query on round-robin partitioned table
Query results saved to table: Test6
        ✓ Test 6 - Round-robin basic: PASSED (5 points)

[Test 7] Point query with different UTC on round-robin
Query results saved to table: Test7
        ✓ Test 7 - Round-robin different UTC: PASSED (5 points)

[Test 8] Verify point query creates table on round-robin
Query results saved to table: Test8
        ✓ Test 8: PASSED (5 points) - Table created successfully on round-robin

[Test 9] Verify consistency between range and round-robin results
Query results saved to table: Test9_range
Query results saved to table: Test9_rrobin
        ✓ Test 9: PASSED (5 points) - Consistent results between partitioning methods

[Test 10] Verify query creates new table correctly
Query results saved to table: Test10
        ✓ Test 10: PASSED (5 points) - New table created successfully

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STEP 2: RANGE QUERY TESTS (50 points)
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--- Range Query on Range Partitioned Table (25 points) ---

[Test 11] Basic range query with valid range
        ✓ Test 11 - Basic range query: PASSED (5 points)
```

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